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NCE Exam Practice Platform — Mauritius

2026

Information & Communication Technology

Grade 9 — National Certificate of Education (NCE)

PREDICTED PAPER 2 — Data Representation, Databases & Cybersecurity

Specimen Practice Paper — Not an Official MES Document

Duration: 1 hour 45 minutes | **Total Marks:** 100

Candidate Name: _____

Class / School: _____

INSTRUCTIONS TO CANDIDATES

1. Write your name and class in the spaces provided above.
2. Answer **ALL** questions in all three sections.
3. Section A: Tick **ONE** box only for each multiple-choice question.
4. Section B: Write your answers in the spaces provided.
5. Section C: Use additional paper if the space provided is insufficient.
6. Calculators are **NOT** permitted.
7. Write clearly in black or blue ink; pencil may be used for diagrams only.

Section	Topic	Marks
A	Multiple Choice Questions	25
B	Short Structured Questions	40
C	Extended / Scenario Questions	35
TOTAL		100

SECTION A — Multiple Choice Questions [25 marks]

For each question, tick (✓) the box next to the **ONE** correct answer. Each question is worth **1 mark**.

1. Which number base does the binary system use? 1
- ☐ A Base 2
- ☐ B Base 8
- ☐ C Base 10
- ☐ D Base 16
2. What is the decimal value of the binary number $1101\ 0110_2$? 1
- ☐ A 210
- ☐ B 214
- ☐ C 206
- ☐ D 218
3. How many different values can be represented using 4 bits? 1
- ☐ A 4
- ☐ B 8
- ☐ C 16
- ☐ D 32
4. The hexadecimal digit F represents which decimal value? 1
- ☐ A 14
- ☐ B 15
- ☐ C 16
- ☐ D 12
5. A sound file is sampled at 44,100 Hz with a bit depth of 16 bits. What do these values control? 1
- ☐ A 44,100 Hz = number of bits per sample; 16 bits = speed of playback
- ☐ B 44,100 Hz = samples taken per second; 16 bits = precision of each sample
- ☐ C 44,100 Hz = file size; 16 bits = channel count
- ☐ D 44,100 Hz = frequency of the highest note; 16 bits = number of tracks

6. In a relational database, a *primary key* is: 1
- ☐ A any field that can contain duplicate values
 - ☐ B a unique identifier for each record in a table
 - ☐ C a field that links to a primary key in another table
 - ☐ D the first column of any database table
7. Which SQL keyword is used to retrieve data from a table? 1
- ☐ A INSERT
 - ☐ B UPDATE
 - ☐ C SELECT
 - ☐ D CREATE
8. A *foreign key* in a database table: 1
- ☐ A must always be unique in its own table
 - ☐ B references the primary key of another table to create a relationship
 - ☐ C stores encrypted data from outside the database
 - ☐ D is a key used to access the database from a foreign computer
9. What does the SQL clause WHERE Age > 18 do? 1
- ☐ A Sorts records by age in ascending order
 - ☐ B Filters records to include only those where Age is greater than 18
 - ☐ C Creates a new column called Age with values over 18
 - ☐ D Deletes all records where Age is greater than 18
10. Which of the following best describes data *validation* in a database? 1
- ☐ A Checking that data entered matches what the user intended
 - ☐ B Ensuring data meets defined rules before it is accepted
 - ☐ C Encrypting all data in the database
 - ☐ D Removing duplicate records from a table
11. A *phishing* attack is best described as: 1
- ☐ A malicious software that self-replicates across a network
 - ☐ B an attempt to trick users into revealing sensitive information via fake emails or websites
 - ☐ C software that monitors user activity without consent
 - ☐ D flooding a server with requests to make it unavailable
12. What is the purpose of encryption in cybersecurity? 1
- ☐ A To compress files so they transmit faster

- ☐ B To convert data into a coded form that only authorised parties can read
- ☐ C To scan files for malware
- ☐ D To verify that software is up to date

13. A firewall in a network is used to:

1

- ☐ A speed up data transfer between devices
- ☐ B monitor and control incoming and outgoing network traffic based on rules
- ☐ C store backup copies of all network data
- ☐ D assign IP addresses to devices

14. Which type of malware disguises itself as a legitimate program to trick a user into installing it?

1

- ☐ A Worm
- ☐ B Spyware
- ☐ C Trojan horse
- ☐ D Adware

15. Two-factor authentication (2FA) improves security by:

1

- ☐ A requiring two different passwords
- ☐ B requiring a password plus a second verification method (e.g. a code sent to a phone)
- ☐ C encrypting the password twice before storing it
- ☐ D allowing two users to share one account securely

16. Which of the following is an example of a strong password?

1

- ☐ A password123
- ☐ B John1990
- ☐ C Tr#9\$mK!2vLq
- ☐ D qwerty

17. The Unicode standard was developed primarily to:

1

- ☐ A replace binary and hexadecimal in computing
- ☐ B represent characters from all world languages using a single standard
- ☐ C compress text files for storage
- ☐ D encrypt text data during transmission

18. Sound is analogue by nature. When recording digitally, a higher *sample rate* means:

1

- ☐ A fewer samples taken, resulting in better quality
- ☐ B more samples taken per second, resulting in better quality but a larger file
- ☐ C the volume of the recording is increased
- ☐ D the bit depth increases automatically

19. In SQL, which command is used to add a new record to a table?

1

- ☐ A ADD INTO
- ☐ B INSERT INTO
- ☐ C PUT INTO
- ☐ D APPEND TO

20. A DDoS (*Distributed Denial of Service*) attack works by:

1

- ☐ A stealing data from a server without the owner knowing
- ☐ B overwhelming a server with requests from many sources, making it unavailable
- ☐ C intercepting messages between two parties
- ☐ D inserting malicious code into a database via a form

21. Which of the following represents pixel colour data in a 24-bit image stored in hexadecimal?

1

- ☐ A #GGHHII
- ☐ B #1A3C6E
- ☐ C 101010
- ☐ D 255, 255, 255

22. Data *verification* in a database means:

1

- ☐ A checking that data follows set rules such as a range or format
- ☐ B confirming that data entered matches the original source (e.g. double entry)
- ☐ C deleting all incorrect records automatically
- ☐ D encrypting data before storage

23. In the SQL query `SELECT * FROM Students WHERE Grade = 9`, what does the asterisk (*) mean?

1

- ☐ A Multiply all values in the Grade column
- ☐ B Select all columns
- ☐ C Select all tables in the database
- ☐ D Select only rows where a column is not empty

24. Ransomware is a type of malware that:

1

- ☐ A displays unwanted advertisements
- ☐ B slows down a computer by using up its resources
- ☐ C encrypts a victim's files and demands payment for the decryption key
- ☐ D records keystrokes and sends them to an attacker

25. Which of the following is the best way for a user to protect personal data on a laptop?

1

- ☐ A Use the same password for all accounts
- ☐ B Never update the operating system
- ☐ C Enable full-disk encryption and use a strong password
- ☐ D Share login details with trusted friends only

SECTION B — Short Structured Questions [40 marks]

Answer **ALL** questions.

B1. Data Representation — Numbers

5

- (a) Convert the binary number $0111\ 1010_2$ to decimal. Show your working.

[2]

- (b) Convert the decimal number 163 to hexadecimal. Show your working.

[2]

- (c) Explain why computers store all data in binary (0s and 1s).[1]

B2. Image Representation

5

- (a) State what *colour depth* means in a bitmap image.[1]

- (b) A bitmap image is 800×600 pixels with a colour depth of 8 bits per pixel. Calculate the uncompressed file size in bytes and in kilobytes. Show your working.

[3]

- (c) State one way to reduce the file size of an image without changing its dimensions.

[1]

B3. Database Tables

5

The table below is part of a school database:

StudentID	FirstName	Surname	Grade	DOB
1001	Priya	Ramkisoorn	9	2012-03-15
1002	Jean-Marc	Laborde	8	2013-07-22
1003	Fatima	Oozeer	9	2012-11-04
1004	Dev	Gobin	7	2014-01-30

(a) Identify the primary key of this table and explain why it is suitable.[2]

(b) Write an SQL query to display all columns for students in Grade 9.[2]

```
SELECT _____  
FROM Students  
WHERE _____;
```

(c) State one validation rule that could be applied to the Grade field.[1]

B4. Database Relationships

5

(a) Explain the difference between a *primary key* and a *foreign key*. [2]

(b) A second table called **Results** has fields: ResultID, StudentID, Subject, Mark. State which field links **Results** to the **Students** table and what type of key it is. [2]

- (c) Give one advantage of using a relational database over a flat-file database. [1]

B5. SQL Queries

5

- (a) Write an SQL query to display the FirstName and Surname of all students whose Surname is 'Gobin'. [2]

- (b) Write an SQL query to display all fields from the Results table for students who scored more than 75 marks, sorted by Mark in descending order. [3]

B6. Cybersecurity Threats

5

- (a) Define the term *malware*. [1]

- (b) Complete the table by naming the type of malware described. [3]

Description	Type of Malware
Replicates itself across a network without user action	
Records keystrokes and sends them to an attacker	
Locks files and demands payment to restore access	

- (c) State one technical measure an organisation can implement to reduce the risk of malware infection. [1]

B7. Encryption

5

- (a) Explain the term *encryption* as used in cybersecurity. [2]

(b) State the difference between *symmetric* and *asymmetric* encryption.[2]

(c) Name one situation where encryption is essential and explain why.[1]

B8. Sound Representation

5

(a) Explain what is meant by *analogue-to-digital conversion* when recording sound.

[2]

(b) A sound clip is recorded with a sample rate of 22,050 Hz, bit depth of 16 bits, for 10 seconds (mono). Calculate the file size in bytes. Show your working.

[2]

(c) State one effect of increasing bit depth on the quality and file size of a recording.

[1]

SECTION C — Extended / Scenario Questions [35 marks]

C1. Scenario: School Examination Database

12

A school uses a database to manage its end-of-year exam results. The database has two tables:

Students(StudentID, FirstName, Surname, Class, DateOfBirth)

Results(ResultID, StudentID, SubjectName, Mark, Grade)

StudentID in the Results table links to the Students table. All marks are out of 100.

- (a) For each table, identify the primary key and state any foreign key. Give a reason for your choices. [4]

- (b) Write an SQL query to display the *FirstName*, *Surname*, and *Mark* for all students who scored **more than 80** in *ICT*. Your query must use a JOIN to combine both tables. [4]

- (c) The school decides to add a validation rule to the Mark field. Suggest a suitable validation rule and explain why it is needed. [2]

- (d) Explain one advantage of using a relational database (with two linked tables) compared to storing all the data in one flat-file spreadsheet. [2]

C2. Scenario: Cybersecurity Incident at a Bank

12

Coastal Bank receives a report that a staff member has clicked on a link in an email claiming to be from the IT department. The link led to a fake login page. The staff member entered their username and password. Shortly after, an unauthorised login to the internal HR system was detected. The bank's security team also notices unusual outbound network traffic.

- (a) Name the type of attack the staff member experienced. Explain how it works. [3]

- (b) Suggest *three* technical measures the bank should implement to prevent a similar attack in the future. For each, give a brief explanation of how it helps. [6]

Measure 1:

Measure 2:

Measure 3:

- (c) The bank encrypts all data stored on its servers. Explain how encryption would limit the damage even if an attacker gained access to the data. [3]

C3. Data Representation — Extended

11

- (a) A character in a text file has ASCII code 72. State what character this is and give its 8-bit binary representation. [2]

- (b) Explain one limitation of the original ASCII standard and how Unicode addresses it. [3]

- (c) A digital photograph at 1920×1080 resolution uses 24-bit colour. Calculate:
(i) The total number of pixels in the image. [1]

- (ii) The uncompressed file size in megabytes (to 2 decimal places). Show full working. [3]

- (d) A JPEG version of the same photo is 2.1 MB. Calculate the compression ratio compared to the uncompressed size. [2]

MARK SCHEME — Paper 2

Award marks for equivalent correct answers. For SQL, accept minor variations in spacing/capitalisation provided the logic is correct.

SECTION A — Multiple Choice Answers

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	B	3	C	4	B	5	B
6	B	7	C	8	B	9	B	10	B
11	B	12	B	13	B	14	C	15	B
16	C	17	B	18	B	19	B	20	B
21	B	22	B	23	B	24	C	25	C

SECTION B — Short Answers

Q	Expected Answer	Marks
B1(a)	$64+32+16+8+2 = 122$ [1 for method, 1 for 122]	2
B1(b)	$163 \div 16 = 10 \text{ r } 3$; so $A3_{16}$ [1 method, 1 correct: A3]	2
B1(c)	Electronic circuits can only be in two states (on/off, high/low voltage), which naturally maps to binary 1 and 0	1
B2(a)	Number of bits used to represent each pixel's colour; determines how many colours are available	1
B2(b)	$800 \times 600 = 480,000$ pixels [1]; $\times 8$ bits = 3,840,000 bits; $\div 8 = 480,000$ bytes [1]; $\div 1024 = 468.75$ KB [1]	3
B2(c)	Reduce colour depth / apply compression (e.g. JPEG) / reduce image quality setting (any one valid answer)	1
B3(a)	StudentID [1]; it is unique for every student so no two records share the same value [1]	2
B3(b)	SELECT * (or all column names) [1]; WHERE Grade = 9 [1]	2
B3(c)	Grade must be between 1 and 9 (range check) / Grade must be an integer / Grade must be present (any valid rule)	1
B4(a)	Primary key: unique identifier for each record in its own table [1]; foreign key: field that references the primary key of another table to create a link [1]	2
B4(b)	StudentID in Results is the foreign key [1]; it links to StudentID (primary key) in Students [1]	2
B4(c)	Reduces data redundancy / avoids repeating student details in every result row / easier to update student info in one place (any one)	1
B5(a)	SELECT FirstName, Surname FROM Students WHERE Surname = 'Gobin'; [1 per correct clause, max 2]	2

B5(b)	SELECT * FROM Results WHERE Mark > 75 ORDER BY Mark DESC; [1 per correct part: SELECT*, WHERE, ORDER BY DESC — 3 marks]	3
B6(a)	Malicious software designed to harm, disrupt, or gain unauthorised access to a computer system	1
B6(b)	Worm [1]; Keylogger [1]; Ransomware [1]	3
B6(c)	Install/update antivirus software / enable firewall / keep OS patched / restrict user privileges (any one valid technical measure)	1
B7(a)	Encryption converts data into a coded (ciphertext) form [1] using an algorithm and key, so only authorised parties with the key can read it [1]	2
B7(b)	Symmetric: uses the same key to encrypt and decrypt [1]; asymmetric: uses a public key to encrypt and a different private key to decrypt [1]	2
B7(c)	Online banking / e-commerce / sending login credentials (any valid situation [1]) — because sensitive financial/personal data could be intercepted [1]	1
B8(a)	The continuous (analogue) sound wave is measured (sampled) at regular time intervals [1]; each sample is converted to a binary number for digital storage [1]	2
B8(b)	$22,050 \times 16 \times 10 = 3,528,000$ bits; $\div 8 = 441,000$ bytes [1 for correct working, 1 for 441,000 bytes]	2
B8(c)	Greater bit depth means more precise amplitude levels, improving quality but increasing file size	1

SECTION C — Extended Answers

Q	Expected Answer / Mark Points	Marks
C1(a)	Students PK: StudentID — unique per student [1]; Results PK: ResultID — unique per result record [1]; FK in Results: StudentID [1]; links to Students.StudentID to identify which student the result belongs to [1]	4
C1(b)	SELECT Students.FirstName, Students.Surname, Results.Mark FROM Students JOIN Results ON Students.StudentID = Results.StudentID WHERE Results.SubjectName = 'ICT' AND Results.Mark > 80; [1 per correct clause: SELECT with correct fields, JOIN with ON, WHERE subject, WHERE mark — 4 marks]	4
C1(c)	Mark must be between 0 and 100 (range check) [1]; prevents invalid marks (e.g. negative or over 100) being entered which would corrupt analysis [1]	2
C1(d)	Reduces data redundancy — student name/class stored once in Students, not repeated for every exam result [1]; easier to maintain — changing a student's name only requires one update [1]	2
C2(a)	Phishing [1]; attacker sends a deceptive email claiming to be from a trusted source [1]; victim clicks a link and enters credentials on a fake website, which the attacker captures [1]	3
C2(b)	Any 3 of: Two-factor authentication — even with stolen password, attacker cannot log in without second factor [2];	6

	Email filtering/spam detection — suspicious links blocked before reaching user [2]; Staff cybersecurity awareness training — employees learn to recognise phishing attempts [2]; Firewall/web filtering — blocks known malicious URLs [2]; Password manager with strong unique passwords — limits damage if one credential is stolen [2] (2 marks per measure: name[1] + explanation[1])	
C2(c)	Even if the attacker downloaded the encrypted data [1], they would only see unreadable ciphertext without the decryption key [1]; the bank's customers' personal and financial data would remain confidential [1]	3
C3(a)	ASCII 72 = 'H' [1]; binary: 01001000 [1]	2
C3(b)	ASCII only has 128 (or 256 extended) characters, insufficient for non-Latin scripts/languages [1]; Unicode (e.g. UTF-8/UTF-16) uses more bits per character [1]; can represent over 1 million characters covering all world scripts, symbols, and emoji [1]	3
C3(c) (i)	$1920 \times 1080 = 2,073,600$ pixels	1
C3(c) (ii)	$2,073,600 \times 3 \text{ bytes} = 6,220,800 \text{ bytes}$ [1]; $\div (1024 \times 1024) = 5.93 \text{ MB}$ [1]; working shown [1]	3
C3(d)	Compression ratio = $5.93 \div 2.1 \approx 2.82:1$ (accept answers in range 2.8:1 – 2.9:1) [1 for correct method, 1 for correct ratio]	2